Field2Sim: Turning In-Field Images into Simulation-Ready Plant Assets

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A row-scale plant asset reconstructed by Field2Sim. From handheld smartphone images, Field2Sim recovers a photorealistic Gaussian representation of a fruit-bearing capsicum row and separates it from the surrounding environment. The resulting plant-only asset preserves the appearance and structural variation of the observed crop and is further decomposed into individual fruit instances for visual simulation and perception.

Abstract

Realistic, semantically annotated plant models are essential for evaluating agricultural robot perception and view planning under representative visual conditions in simulation, but creating such assets from real crops remains difficult. Existing approaches often trade annotation quality for visual realism or require specialized capture equipment. We introduce Field2Sim, a modular pipeline that converts handheld smartphone images of crop rows into photorealistic Gaussian Splatting reconstructions with separated plants and individually identified fruits. Field2Sim selects its reconstruction backbone based on downstream fruit-segmentation performance, transfers 2D semantic masks into 3D without additional segmentation training, and introduces a novel two-pass fruit-instance identification method that refines geometric clusters using multi-view 2D instance evidence. We evaluate Field2Sim on five capsicum rows in a commercial greenhouse containing 1,722 annotated fruits, where the two-pass method achieves a 3D panoptic quality of 0.954, compared with 0.808 for geometric clustering alone, and reduces fruit-count error from 20.0% to 1.5%. We publicly release a dataset of the five reconstructions with their plant and fruit annotations, together with our code, to support simulation-based research on agricultural robot perception and view planning.

1. Introduction

Agricultural automation, from field robots to networked camera and sensor systems, increasingly relies on simulated environments to train and validate its perception and decision-making components [29]. These environments can provide perception models with labeled training images, such as segmentation masks and depth maps, without requiring manual annotation [22]. They also allow perception-driven robot policies, such as view planning or navigation, and different sensor configurations to be evalu-

ated in closed loop prior to deployment in real fields, including from viewpoints that need not have been physically captured [10]. Since plants make up most of what these systems observe, the utility of such environments depends on the visual realism of their 3D plant models and on the semantic information those models provide. Existing approaches to generating crop assets typically achieve one of these properties at the expense of the other. Procedural systems and artist-designed meshes provide complete semantic labels by construction, along with controllable geometry and straightforward scene randomization [3, 4]. However, their realism is limited by the complexity of the underlying models or the manual effort invested in their design, and the resulting assets can remain visibly synthetic. Reconstruction from real plants addresses this limitation. Terrestrial laser scanners, structured-light sensors, and X-ray tomography can recover plant geometry with high accuracy [8, 28], but require specialized acquisition equipment and do not inherently provide photorealistic appearance or semantic labels. Image-based reconstruction using radiance fields or Gaussian splatting lowers the barrier to acquisition by relying on ordinary cameras rather than specialized sensing equipment while retaining detailed visual appearance. In plant applications, however, these representations have largely been developed for phenotyping, with semantic or instance information used primarily to support measurements and analysis rather than to produce reusable, semantically annotated plant assets [23, 24, 35]. As such, obtaining plant assets that are both photorealistic and semantically annotated from real crops, using only commodity cameras, remains difficult.

To fill this gap, we propose **Field2Sim**, a modular pipeline built around Gaussian Splatting [12] that generates simulation-ready plant assets from images captured in fields and greenhouses. We define a plant asset as *simulation-ready* if it (i) preserves a photorealistic appearance, (ii) encodes semantic information, and (iii) can be readily imported into existing simulation frameworks. Field2Sim leverages recent advances in vision foundation models, such as SAM 3 [6], to generate 2D semantic masks for the captured images; these masks are then lifted onto the 3D Gaussian primitives using training-free techniques. Importantly, we validate our choice of Gaussian Splatting training method through its downstream semantic segmentation performance, since standard photometric metrics (e.g., PSNR, SSIM, and LPIPS) do not capture the particular challenges posed by agricultural scenes. Finally, we introduce a novel two-pass instance segmentation method that addresses the difficulty of assigning individual instance labels to densely clustered and partially occluded fruits, and we compare it against geometry-based approaches.

In summary, our main contributions are:

- **Field2Sim**, a modular pipeline that transforms casually captured greenhouse images into photorealistic, semantically decomposed plant and fruit assets, without specialized sensing equipment or scene-specific fine-tuning.
- A **controlled evaluation** of the pipeline's main design choices, comparing five Gaussian Splatting reconstruction strategies by downstream semantic segmentation, and a novel **two-pass fruit instance segmentation** method evaluated against geometric clustering baselines.
- A **curated dataset** of five capsicum greenhouse-row reconstructions with semantic plant representations and individual fruit assets, publicly released[1] to support perception and planning research in agricultural vision and robotics.

2. Related Work

Radiance Field Reconstruction. Neural Radiance Fields (NeRFs) [25] demonstrated photorealistic novel-view synthesis by optimizing a continuous scene representation through differentiable volume rendering. 3D Gaussian Splatting (3DGS) [12] subsequently introduced an explicit representation based on anisotropic Gaussian primitives, enabling faster optimization, real-time rendering, and direct manipulation of the reconstructed scene. Several extensions address limitations of its original optimization strategy: AbsGS [33] mitigates gradient cancellation during densification, Taming 3DGS [20] controls primitive allocation under a fixed budget, and 3DGS-MCMC [15] replaces heuristic densification with a sampling-based formulation. Complementary approaches improve scale-aware rendering [34] or support large-scale scenes through hierarchical optimization [13]. While these methods perform strongly on standard benchmarks, their relative behavior remains underexplored for row-grown crops, which combine repeated appearance, thin structures, severe occlusion, and non-uniform view coverage.

3D Scene Understanding. While 3D scene understanding has traditionally operated on point clouds and meshes, differentiable rendering enables semantic information from pretrained 2D models to be transferred directly into radiance-field representations. LERF [14] distills CLIP features into a NeRF, while Feature 3DGS [36] and LangSplat [26] attach distilled features to Gaussian primitives for efficient open-vocabulary querying. Rather than learning an additional feature field, FlashSplat [30] directly solves the assignment of 2D masks to Gaussians, whereas LUDVIG [21] uses learning-free feature aggregation and graph diffusion. Instance-level understanding remains more involved because 2D instance identities are generally inconsistent across views. Methods such as Panoptic Lifting [31], GARField [16], and PanopticSplatting [32] address this through learned instance labels, affinity fields, or object queries. For our application, we instead employ a post-

[1] https://dtabaa.github.io/field2sim

hoc procedure that combines geometric grouping with 2D instance evidence to recover individual fruits.

Plant Reconstruction and Simulation. Agricultural simulation has traditionally relied on procedural plant models. Helios [3] combines organ-level geometry with biophysical simulation, while CropCraft [4] generates configurable crop fields and annotations for agricultural perception. These systems provide controllable geometry and labels, but their structural diversity and appearance remain governed by authored models and procedural rules.

Image-based methods preserve observed crop morphology, although most target specific phenotyping outputs. Choi *et al*. [7] extract NeRF geometry and apply skeletonization, surface reconstruction, and geometric fitting to estimate plant traits. FruitNeRF [23] samples a semantic NeRF into a fruit point cloud for clustering, while FruitNeRF++ [24] introduces contrastive instance embeddings for fruit-agnostic counting. CountingFruit [18] instead embeds language features into 3D Gaussians and clusters prompt-selected samples, and Wheat3DGS [35] associates detector- and SAM-derived masks with Gaussians to isolate and measure wheat heads. These methods primarily produce task-specific counts or measurements rather than reusable plant and fruit components. GrowSplat [2] and Botany-Bot [1] reconstruct richer plant digital twins, but rely on fixed multi-camera or controlled robotic acquisition. In contrast, Field2Sim targets simulation-ready reconstruction from more accessible data: it reconstructs greenhouse rows from handheld images, then applies post-hoc semantic lifting and instance separation to obtain plant- and fruit-level assets without dedicated scanning infrastructure.

3. Field2Sim

We present Field2Sim, a framework for generating simulation-ready agricultural assets from images captured with commodity hardware. This section outlines the pipeline (Sec. 3.1) and the principles behind its design (Sec. 3.2).

3.1. System Overview

Given a set of handheld RGB images of a crop row, Field2Sim constructs a photorealistic 3D Gaussian representation with plant-level semantic labels and individual fruit instances. As illustrated in Fig. 2, we first estimate camera poses and sparse scene geometry using Structure-from-Motion (SfM), then use these estimates to optimize a 3D Gaussian Splatting (3DGS) representation. The reconstructed scene is subsequently aligned and spatially cropped to remove distant artifacts. We apply SAM 3 to obtain semantic plant masks and fruit instance masks, which LUDVIG lifts into 3D in two successive passes to isolate plant and fruit Gaussians. Individual fruits are then identified through a two-stage procedure: an initial decomposition into 3D connected components, followed by selective refinement of components containing evidence of multiple fruit instances in the 2D masks. The resulting representation preserves the appearance of the captured crop row while exposing individual plants and fruits for downstream simulation tasks.

3.2. Design Principles

We follow three design principles that guide both the selection of existing components and the design of our own methods within the framework:

- **Modularity.** Each stage makes minimal assumptions about adjacent components, allowing individual methods to be replaced as stronger alternatives become available.
- **No fine-tuning.** We favor methods that operate out of the box, avoiding dataset- or scene-specific training and additional annotations.
- **Scalability.** We prioritize methods that require limited manual intervention and remain practical as the number of images, plants, and scenes grows.

The following sections detail the pipeline in order. We evaluate the component choices and resulting assets in Sec. 7.

4. Scene Reconstruction

4.1. Sparse Geometry Estimation

Our framework uses Gaussian Splatting to obtain photorealistic representations of agricultural plants, requiring camera poses and a sparse 3D point cloud to initialize the Gaussian primitives. We recover both using an incremental Structure-from-Motion (SfM) pipeline [27], which establishes feature correspondences across images and jointly estimates camera poses and sparse scene geometry. Since reconstruction is performed offline, we prioritize reconstruction robustness over computational efficiency.

The strong visual self-similarity between plants and repeated structures can produce incorrect correspondences between spatially distinct regions, degrading image registration and pose estimation. We therefore restrict feature matching to a sliding window of 50 images around each image in the acquisition sequence. Unlike exhaustive or vocabulary-tree-based matching, this constraint uses the acquisition order to reduce geometrically implausible correspondences between visually similar plant regions.

4.2. Radiance Field Optimization

Given the recovered camera poses and sparse geometry, we optimize the scene as a set of 3D Gaussian primitives. We use 3DGS because its representation is explicit, portable, and can be directly rendered and edited by downstream stages. We therefore avoid variants that introduce additional

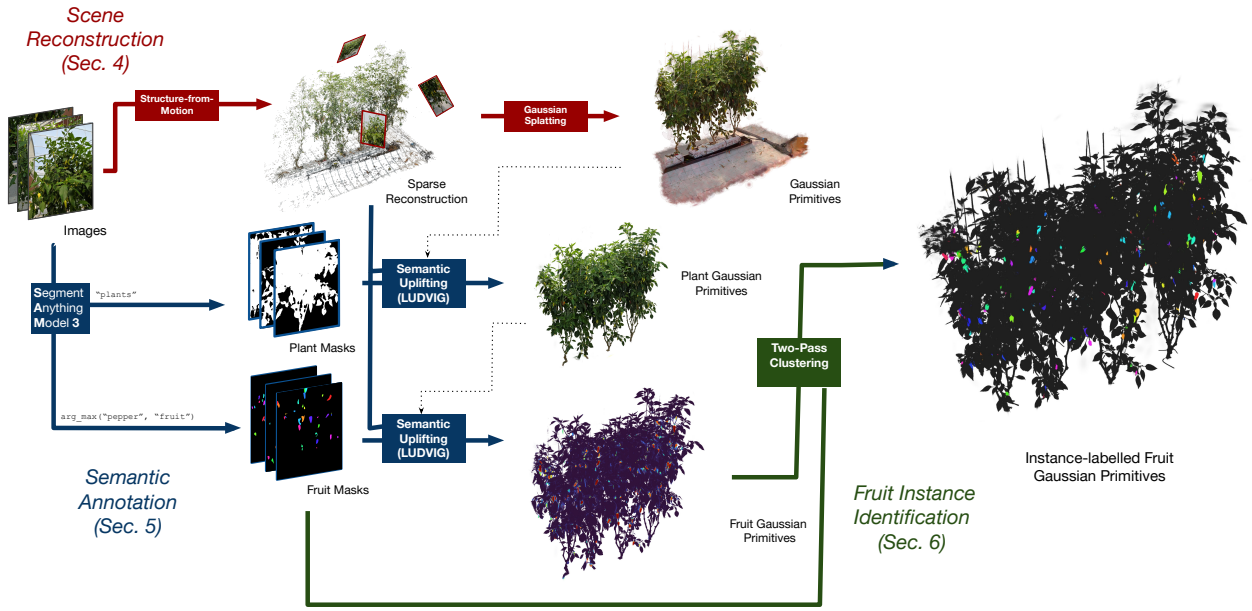


Figure 2. **Field2Sim pipeline.** Handheld RGB images are reconstructed into 3D Gaussian primitives, semantically decomposed into plant and fruit representations, and processed to identify individual fruit instances. Dotted arrows indicate reconstruction information reused during semantic lifting.

neural parameters, such as Scaffold-GS [19], which would make the exported representation less self-contained.

Based on the empirical comparison presented in Sec. 7.2, we use the chunk-level optimizer of Hierarchical 3DGS [13]. Its densification strategy replaces the average screen-space positional gradient used by the original 3DGS with the maximum gradient observed over the densification window. Let $\tilde{\boldsymbol{\mu}}_i^{(t)}$ denote the image-space mean of Gaussian i at optimization step t, and let $\mathcal{T}_i$ contain the steps during which it is observed. The corresponding densification statistic is

$$g_i = \max_{t \in \mathcal{T}_i} \left| \frac{\partial \mathcal{L}_t}{\partial \tilde{\boldsymbol{\mu}}_i^{(t)}} \right|_2 , \tag{1}$$

with Gaussian i densified when $g_i > \tau_{\text{dens}}$. This prevents strong refinement signals from a small number of informative views from being diluted by weaker observations, which is useful under the uneven view coverage and heavy occlusion present in our captures. We use only this chunk-level optimization and do not construct the global hierarchy, leaving a standard set of Gaussian primitives for subsequent processing.

5. Semantic Annotation

Several methods augment Gaussian splatting with semantic information [26, 30, 36]. However, many require optimizing additional scene-specific features or coupling semantic lifting with reconstruction. To preserve the modularity of Field2Sim, we instead use LUDVIG [21], which transfers 2D masks to an already optimized Gaussian representation without further training or modifying its parameters.

We apply semantic lifting sequentially. Plant masks first separate the crop from the surrounding greenhouse structures. Fruit masks are then lifted onto the resulting plant representation to isolate the fruit Gaussians used for instance identification. Although we focus on peppers, the same procedure can be applied to other visible plant organs, such as flowers or leaves, by changing the 2D prompt.

5.1. 2D Semantic Mask Generation

We use SAM 3 [6] because it performs well in our setting without fine-tuning. For every input image, we generate a semantic plant mask and both semantic and instance-level fruit masks. The semantic masks provide supervision for 3D lifting, whereas the instance masks are retained as multi-view evidence for the fruit-separation stage described in Sec. 6.

Since SAM 3 is text-prompted, its predictions depend on the noun used to describe the target. The prompt `plants` is used for plant isolation across all scenes. For fruit segmentation, we compare a small set of candidate nouns on sampled frames from each scene and select the prompt producing the strongest detection coverage.

5.2. 3D Semantic Uplifting

LUDVIG interprets semantic lifting as an inverse rendering operation. Let $q = (d, p)$ denote pixel p in view d, and let $\mathcal{S}_i$ contain all such observations influenced by Gaussian i. Its uplifted semantic score is

$$s_i = \frac{\sum_{q \in \mathcal{S}_i} w_{iq} M_q}{\sum_{q \in \mathcal{S}_i} w_{iq}}, \tag{2}$$

where M_q is the corresponding binary mask value and w_{iq} is the alpha-blending contribution of Gaussian i to that pixel. Thus, s_i measures how consistently the Gaussian contributes to foreground pixels across the observed views.

We first uplift the plant masks onto the complete scene and retain Gaussians whose scores exceed $\tau_{\text{plant}} = 0.3$, producing a plant-only representation. We then repeat the operation on this filtered representation using the semantic pepper masks and $\tau_{\text{fruit}} = 0.5$. The result is a fruit-only set of Gaussians, while their geometry, appearance, opacity, and other reconstruction parameters remain unchanged. These Gaussians form the input to the instance-identification procedure described in Sec. 6.

6. Fruit Instance Identification

Agricultural simulation tasks such as fruit counting and the evaluation of view-planning coverage require instance masks that distinguish individual fruits. Separating fruit instances also allows plant configurations to be randomized by removing, duplicating, or repositioning fruits (see Sec. 7.4). However, tightly clustered or touching fruits can form a single connected structure in 3D, making it difficult to recover individual instances from the fruit-level semantic representation. Occlusion and incomplete observations further limit the reliability of shape- and size-based separation. We therefore propose a two-pass procedure that first groups fruit Gaussians by geometric proximity, then uses multi-view instance masks to identify and refine components containing multiple fruits.

Initial Pass. We first construct a proximity graph over the segmented fruit Gaussians and extract its connected components. Since Gaussian primitives encode both position and spatial extent, we use a symmetric Mahalanobis distance rather than Euclidean distance. For a pair of Gaussians i and j, with means $\boldsymbol{\mu}_i$ and $\boldsymbol{\mu}_j$ and covariances $\boldsymbol{\Sigma}_i$ and $\boldsymbol{\Sigma}_j$, we define

$$d(i,j) = \sqrt{(\boldsymbol{\mu}_j - \boldsymbol{\mu}_i)^\top \left(\frac{\boldsymbol{\Sigma}_i + \boldsymbol{\Sigma}_j}{2} + \epsilon \mathbf{I} \right)^{-1} (\boldsymbol{\mu}_j - \boldsymbol{\mu}_i)}. \tag{3}$$

To avoid considering all Gaussian pairs, we first generate candidate neighbours within a Euclidean radius determined by the 99th percentile of the maximum Gaussian scales, and retain an edge whenever $d(i,j) < \tau$, where τ controls the permissiveness of the initial geometric clustering.

Further Clustering. Because we already obtain per-image fruit instance masks from SAM 3, we use them to identify components that potentially over-aggregate multiple fruits. For each view, we render the full scene and recover the standard 3DGS contribution $w_{i,p} = \alpha_{i,p} T_{i,p}$ of Gaussian i to pixel p. This naturally accounts for occlusion, as Gaussians hidden by other scene elements have negligible contribution.

Let $\mathcal{G}_k$ denote the Gaussians belonging to component k, and let Ω_m denote the set of pixels belonging to instance mask m. We measure the association between component k and mask m by accumulating their contributions,

$$W_{k,m} = \sum_{i \in \mathcal{G}_k} \sum_{p \in \Omega_m} w_{i,p}. \tag{4}$$

We then normalize these associations over all foreground masks,

$$\hat{W}_{k,m} = \frac{W_{k,m}}{\sum_{m'} W_{k,m'}}, \tag{5}$$

such that $\hat{W}_{k,m}$ represents the fraction of the visible component associated with instance m. A view provides evidence that component k contains multiple fruits when

$$\left| \{ m : \hat{W}_{k,m} \geq \gamma \} \right| \geq 2, \tag{6}$$

where γ is the minimum fraction of the component's foreground contribution that a mask must explain to count as a distinct instance.

Finally, let $\mathcal{V}_k$ be the set of views satisfying this criterion for component k. We flag the component for further clustering when $|\mathcal{V}_k| \geq \nu$, where ν is the minimum number of views that must support the split. This multi-view requirement reduces sensitivity to isolated 2D segmentation errors.

For each flagged component, we select the views in $\mathcal{V}_k$ and run SAM 3 in sequence mode, obtaining consistent instance identities across them. We then uplift these tracked masks onto the component Gaussians using the procedure described in Sec. 5.2. If $F_{i,o}$ denotes the resulting association between Gaussian i and tracked instance o, we assign

$$\ell_i = \arg\max_o F_{i,o}. \tag{7}$$

Gaussians receiving insufficient evidence inherit the label of their nearest assigned neighbour, and small resulting fragments are similarly reabsorbed. The component is split only if at least two valid sub-instances remain after this assignment; otherwise, it is left unchanged.

We perform this refinement independently for each flagged component rather than tracking all fruits jointly, substantially reducing the memory requirements of SAM 3 as the number of simultaneously tracked objects increases.

Table 1. Statistics of the greenhouse scenes used for evaluation. All scenes contain sweet pepper (*Capsicum annuum* L.).

	Capture	Scene characteristics		
Scene	Images	Plants	Fruits†	Row length† (m)
Orange-fruited scenes				
ORANGE-L1	1944	15	438	6.4
ORANGE-L2	3681	15	690	6.5
Red-fruited scenes				
RED-L1	4087	15	250	6.4
RED-S1	1548	6	79	2.5
Yellow-fruited scenes				
YELLOW-S1	875	4	265	1.6
Total	**12135**	**55**	**1722**	**23.4**

†Approximate value.

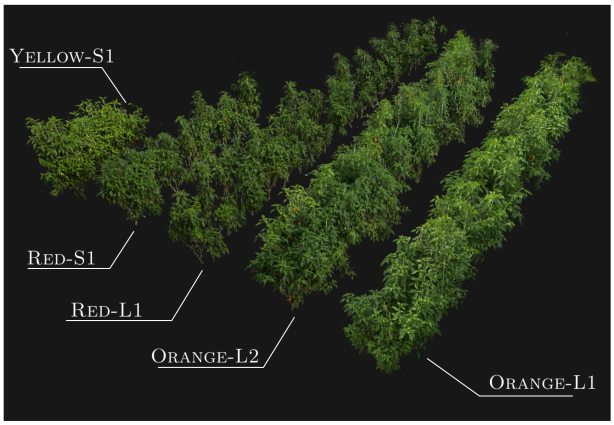


Figure 3. **Overview of the evaluation scenes.** Plant-only Gaussian reconstructions of the five capsicum greenhouse-row segments used throughout our experiments. The collection includes short and long rows with orange-, red-, and yellow-fruited plants.

7. Experiments

7.1. Experimental Setup

Dataset and image acquisition. We evaluate Field2Sim on five segments from different planting rows in a commercial capsicum greenhouse. Two segments contain orange-fruited plants (ORANGE-L1 and ORANGE-L2), two contain red-fruited plants (RED-L1 and RED-S1), and one contains yellow-fruited plants (YELLOW-S1). We capture the images with a Samsung Galaxy S25 Ultra using OpenCamera [11], which allows us to disable the image enhancement and post-processing typically applied by the default camera application. Table 1 summarizes the capture and scene statistics, and Fig. 3 shows the corresponding plant-only reconstructions.

Reconstruction setup. Before reconstruction, we resize all images to 1600×1200 pixels to balance reconstruction quality against the higher memory and runtime requirements of processing the original images. We use COLMAP [27] to recover camera poses and sparse scene geometry through structure from motion, then align the reconstruction. We run all experiments on a workstation equipped with an NVIDIA RTX A6000 GPU.

Segmentation settings. Unless otherwise stated, we use the same segmentation settings across all scenes, with a scene-specific prompt for fruit segmentation. In the prompt-selection experiments reported in the Supplementary Material, *fruit* performs best for ORANGE-L1 and ORANGE-L2, while *pepper* performs best for RED-L1, RED-S1, and YELLOW-S1.

Fruit instance identification. In the first pass, we set the proximity cutoff to $\tau = 4.0$ Mahalanobis units and discard connected components with fewer than 50 Gaussians. In the second pass, we set the minimum mask support to $\gamma = 0.25$ and require evidence from at least three views ($\nu = 3$) to flag a component for refinement. For each flagged component, we track 12 frames and retain tracked identities that cover at least 20% of the component's projected support. After splitting, we merge fragments with fewer than 50 Gaussians into their nearest sibling, preserving all fruit Gaussians that enter the second pass.

7.2. Radiance Field Reconstruction

We compare five 3DGS variants: the original 3DGS, AbsGS, Mip-Splatting, Taming 3DGS, and Hierarchical 3DGS. For the latter, we evaluate only its chunk-level reconstruction stage and do not construct the global hierarchy. All methods are initialized from the same SfM reconstruction and optimized for 30,000 iterations using their published settings, except where noted. We disable exposure optimization for 3DGS and Hierarchical 3DGS, and use a budget multiplier of 20 for Taming 3DGS. To ensure a consistent evaluation, we select one eighth of the images in each scene by farthest-point sampling over the camera centers and use the same held-out views and metric implementation for every method. We report the unweighted mean of the five per-scene scores.

Photometric reconstruction quality. Table 2 reports PSNR, SSIM, and LPIPS, which measure pixelwise fidelity, structural similarity, and perceptual similarity, respectively. The metrics do not agree on a single best method. Taming 3DGS achieves the highest PSNR on all five scenes and the highest mean PSNR and SSIM (21.79 dB and 0.704), whereas Hierarchical 3DGS achieves the lowest mean LPIPS (0.262) and ranks first on LPIPS in three scenes. Within each scene, the PSNR gap between the best and worst methods ranges from 1.09 to 1.70 dB.

Table 2. **Gaussian-splatting reconstruction.** Mean novel-view synthesis performance across five scenes. ■ / ■ / ■ highlight the best, second-, and third-best results. † denotes the chunk-level optimizer of Hierarchical 3DGS.

Method	PSNR ↑	SSIM ↑	LPIPS ↓
3DGS [12]	21.49	0.697	0.264
AbsGS [33]	20.54	0.658	0.281
Mip-Splatting [34]	21.16	0.683	0.282
Taming 3DGS [20]	21.79	0.704	0.265
Hierarchical 3DGS† [13]	20.72	0.688	0.262

Downstream segmentation transfer. Novel-view metrics do not measure whether a rendered scene preserves the fruit evidence required by later perception stages. We therefore run SAM 3 with the same scene-specific prompt on each held-out photograph and its corresponding render. The photograph masks serve as the reference, so this evaluation measures agreement with SAM 3 on real imagery rather than absolute segmentation accuracy. We score only views containing at least one reference instance, retaining renders with no detections as complete misses.

Table 3 reports mask average precision (AP), instance F1, Panoptic Quality (PQ), and the missed-view rate. AP summarizes precision–recall performance, while F1 balances instance precision and recall. PQ [17] combines detection and delineation through $\mathrm{PQ} = \mathrm{SQ} \times \mathrm{RQ}$. Segmentation quality (SQ) is the mean intersection-over-union (IoU) of matched masks, while recognition quality (RQ) is an instance F1 score that penalizes missed reference instances and unmatched render detections. Within this evaluation, Hierarchical 3DGS achieves the strongest measured result for every downstream metric, reaching 0.507 mask AP, 0.655 PQ, and 0.754 F1, while reducing the missed-view rate to 0.155. It leads on four of the five scenes, with the largest margin on RED-L1, where it obtains 0.500 AP compared with 0.355 for the next-best method. Across the five methods, mean SQ ranges narrowly from 0.862 to 0.868, indicating similar mask overlap among matched instances. The principal difference is therefore whether fruit instances are detected, rather than the overlap of successfully matched masks.

The downstream ranking also differs from the photometric one. Taming 3DGS leads PSNR and SSIM but obtains 0.391 mask AP, whereas Hierarchical 3DGS achieves the strongest measured downstream agreement. Higher PSNR and SSIM therefore do not necessarily correspond to better segmentation agreement within this evaluation. Given its lowest mean LPIPS and strongest measured downstream agreement, we use the chunk-level Hierarchical 3DGS optimizer for the remaining Field2Sim experiments.

Table 3. **Downstream fruit-segmentation transfer.** Mean agreement between SAM 3 instance masks from held-out photographs and corresponding renders, using photograph masks as references. $\mathrm{AP}_{50:95}$ denotes mask average precision over IoU thresholds from 0.50 to 0.95; $\mathrm{F1}_{50}$ denotes instance F1 at IoU 0.5. "Miss" is the fraction of scored views with no render-side detection. Symbols and ranking colors follow Table 2.

Method	$\mathrm{AP}_{50:95}$ ↑	PQ↑	$\mathrm{F1}_{50}$ ↑	Miss↓
3DGS [12]	0.400	0.582	0.673	0.235
AbsGS [33]	0.426	0.606	0.702	0.212
Mip-Splatting [34]	0.392	0.566	0.652	0.275
Taming 3DGS [20]	0.391	0.569	0.655	0.270
Hierarchical 3DGS† [13]	0.507	0.655	0.754	0.155

7.3. Fruit Instance Identification

We evaluate fruit instance identification on 1,722 fruit instances comprising 363,844 Gaussian primitives across the five scenes. Ground-truth instance labels were manually created for evaluation. All methods are evaluated on the same reference fruit Gaussians, allowing predicted and reference instances to be compared through exact set intersections.

We compare our complete two-pass method against DBSCAN [9] and HDBSCAN [5], as well as an ablation containing only the first geometric pass. All methods use a minimum instance size of 50 Gaussians, with baseline distance parameters normalized by the median distance to the tenth-nearest fruit Gaussian in each scene. We report Panoptic Quality (PQ) at an IoU threshold of 0.5 and its segmentation and recognition components (SQ and RQ). We also report Adjusted Rand Index (ARI), which measures agreement between predicted and reference instance assignments over pairs of Gaussians, adjusted for chance. Absolute relative count error measures the absolute difference between predicted and reference instance counts as a percentage of the reference count.

Quantitative results. Table 4 reports the unweighted mean over the five scenes. The complete method reaches 0.954 PQ and 0.963 ARI, compared with 0.808 PQ and 0.789 ARI for the geometric pass alone. The second pass therefore improves PQ by 0.146 and reduces mean absolute relative count error from 20.0% to 1.5%. Across all five scenes, the number of reference instances matched to a prediction increases from 1,354 to 1,667 out of 1,722, while unmatched predictions increase from 42 to 51. The complete method achieves PQ scores between 0.943 and 0.976 across scenes.

The geometric pass achieves PQ comparable to the clustering baselines, with 0.808 compared with 0.785 for HDBSCAN and 0.774 for DBSCAN. Both density-based methods under-count in every scene, with mean absolute relative count errors of 20.5% and 12.4%, respectively. To-

Table 4. **Fruit instance identification.** Unweighted means over five scenes. Error denotes absolute relative instance-count error. Best results are shown in bold.

Method	PQ ↑	SQ ↑	RQ ↑	ARI ↑	Error (%) ↓
DBSCAN	0.774	0.904	0.857	0.591	12.4
HDBSCAN	0.785	0.929	0.845	0.765	20.5
Ours (Geometric)	0.808	0.944	0.856	0.789	20.0
Ours (Two-pass)	**0.954**	**0.985**	**0.969**	**0.963**	**1.5**

gether with the geometric-pass ablation, these results indicate that the image-driven refinement provides the main improvement in instance identification.

7.4. Practical Utility

Field2Sim produces reusable visual simulation assets from handheld smartphone images without specialized capture equipment. Across the five captures, a single smartphone was sufficient to reconstruct 23.4 m of greenhouse rows containing approximately 55 plants and 1,722 fruits. Each scene contains a conventional 3DGS model, plant and fruit subsets, and individual fruit labels. These assets can be rendered using standard Gaussian rasterizers without additional neural parameters.

Separating individual fruits allows scene configurations to be varied without repeating capture or reconstruction. Figure 4 demonstrates fruit removal and recoloring from three viewpoints. Removing different subsets of fruits produces many configurations with different counts and spatial distributions while preserving the underlying plant structure. Recoloring changes the visual cues associated with ripeness, allowing scenarios with different distributions of visually ripe target fruits. Fruits can also be duplicated or repositioned to modify their arrangement. Because these edits are applied to the shared 3D representation, each configuration is consistent across rendered viewpoints by construction.

The resulting configurations support fruit-counting and segmentation evaluation, as well as information-gathering and view-planning tasks that seek to discover fruits or inspect targets with particular appearance attributes. RGB images, depth maps, and corresponding semantic or instance masks can be rendered from camera poses within the reconstructed view support. This allows camera placement and perception-driven view selection to be evaluated across controlled scene variants without repeatedly accessing the greenhouse. Since annotation is performed post hoc, updated segmentation models or labels for additional plant organs can also be incorporated without reconstructing the scene again.

The assets are intended for visual simulation and perception. They do not provide watertight collision geometry, biomechanical properties, or plant dynamics, and therefore do not yet support contact-rich tasks such as harvesting or leaf manipulation, which we leave to future work.

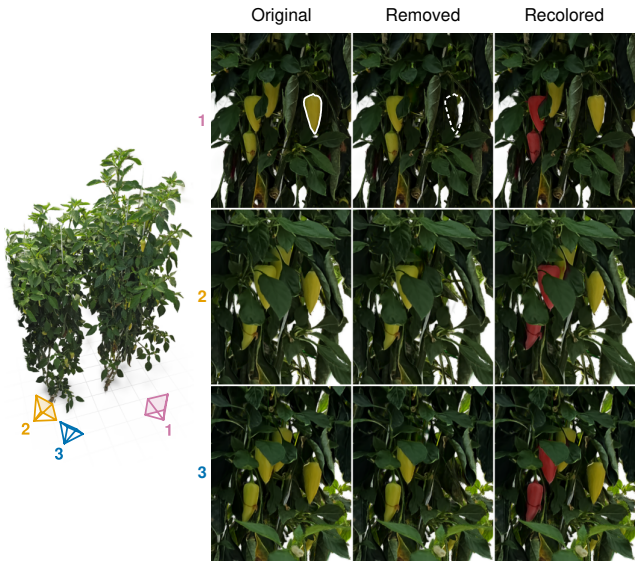


Figure 4. **Instance-level editing for visual simulation.** Left shows the reconstructed plant asset and three camera viewpoints. Right shows the original, fruit-removal, and recolored configurations from each viewpoint. Solid and dashed outlines in the first row mark an example fruit and its location after removal. Removal varies fruit count and spatial distribution, while recoloring varies visual cues associated with ripeness.

8. Conclusions

Field2Sim demonstrates that handheld smartphone images can yield photorealistic plant assets with individual fruit labels for visual simulation. Our evaluation highlights two considerations for building such assets: reconstruction quality should be assessed through downstream perception as well as image similarity, and multi-view instance evidence helps separate neighbouring fruits that geometric clustering alone leaves merged. Together, these findings support combining established reconstruction and segmentation methods with targeted instance refinement to produce reusable, editable crop assets.

The resulting assets provide a basis for controlled studies of fruit perception and view planning, with fruit configurations that can be varied without additional capture. Our evaluation is limited to static capsicum rows from one greenhouse; extending it to other crops and testing the assets in closed-loop robotic tasks are the next steps toward establishing their broader utility.

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